INCH-POUND

MIL-DTL-43176D
15 September, 2000
SUPERSEDING
MIL-S-43176C
7 October, 1987

DETAILED SPECIFICATION

SCREEN, LATRINE, FIRE, WATER WEATHER AND MILDEW RESISTANT TREATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

*1.1 Scope. This specification covers requirements for a latrine screen made of Camouflage Green 483 polyester duck, that is fire, water, weather and mildew resistant treated. A cover is provided to protect the screen during storage and transportation.

2. **APPLICABLE DOCUMENTS**

*2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

*2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions, clarifications) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CNR, 700 Robbins Ave., Philadelphia, PA 19111-5096 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

A-A-50057	- Rope, Fibrous, Tent-Lay
V-T-285	- Thread, Polyester
DDD-L-20	- Label: For Clothing, Equipage and Tentage (General Use)

MILITARY

MIL-P-549	- Poles, Tent, Upright and Ridge
MIL-L-1709	- Lines, Tent
MIL-S-1734	- Slips, Tent Line
MIL-P-2383	- Pins, Tent, Wood
MIL-DTL-20696	- Cloth, Coated, Waterproof
MIL-R-30500	- Rope, Polyester
MIL-DTL-44103	- Cloth, Duck, Fire Water & Weather Resistant

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY BIOLOGICAL & CHEMICAL COMMAND

Department of the Army Field Manual FM 10-16 – General Repair of Tents, Canvas and Webbing

(Copies of drawings are available from the U. S. Army Biological Chemical Command, ATTN: AMSSB-REO-S(N), NATICK, MA 01760-5017.)

DRAWINGS

U.S. ARMY BIOLOGICAL & CHEMICAL COMMAND

5-4-821	- Screen, Latrine, FWWMR, O.D.; Illustration and details
5-4-822	- Cover
5-4-823	- Erection Instructions

(Copies of drawings are available from the U. S. Army Biological Chemical Command, ATTN: AMSSB-REO-S(N), NATICK, MA 01760-5017.)

*2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA

NASM16491 - Grommet, Metallic, General Specification For

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(Applications for copies of referenced documents should be addressed to the Aerospace Industries Association of America, Inc., 1250 Eye Street, N. W. Washington, DC 20005-3924.)

*2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

*3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.2).

3.2 Materials and components. The materials (see 6.5) and components shall be as specified on the applicable drawings and as specified herein.

*3.2.1 Cloth.

*3.2.1.1 Cloth, duck. The cloth for the screen shall conform to class 1, grade B of MIL-PRF-44103.

*3.2.1.2 Cloth, coated. The cloth for the cover shall conform to type II, class 2, shade Camouflage Green 483 of MIL-PRF-20696.

3.2.2 Thread. The thread shall conform to type I, class 1 subclass A or B, shade S-1 (CA 66022) of V-T-285. The dyed thread shall show colorfastness to weathering equal to or better than the standard shade sample (see 6.3). If no standard shade sample is available the thread shall show good colorfastness to weathering. Thread size shall be as follows:

a. For all stitching except stitching indicated by “Z” on drawings size F for needle and bobbin.
b. For stitching indicated by “Z” on drawing size F for needle and size E for looper.

*3.2.3 Lines, tent. The tent lines shall conform to types VI, XV, XVIII and XXI of MIL-L-1709. As an alternate, the following substitute ropes are acceptable for use: ¼ or 5/16 inch diameter rope conforming to A-A-50057. Polyester, Camouflage Green 483 of MIL-R-30500 or three strand, dual fiber polyester/polypropylene (commercial).

*3.2.4 Grommets, metallic. Grommets shall conform to type III, class 3, size No. 4 of NASM16491.

3.2.5 Slips, tent line. Tent line slips shall conform to type II of MIL-S-1734.

3.2.6 Pins, tent, wood. When required, (see 6.2), the tent pins shall be size 1, conforming to MIL-P-2383.

3.2.7 Poles, tent, upright and ridge. When required (see 6.2), the tent poles shall conform to MIL-P-549.

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3.3 Construction. The construction of the latrine screen shall conform to the drawings listed in 2.2.2 and as specified herein.

3.3.1 Stitching, machine.

*3.3.1.1 Types. All stitch types, shall conform to ASTM-D-6193 as follows:

a. For all stitching except stitching indicated by "Z" on drawings – type 301, 5 to 7 stitches per inch.
b. For stitching indicated by "Z" on drawings – type 401, 5 to 7 stitches per inch. Chain portion of stitching shall not appear on outside of screen or cover.

3.3.1.2 Stitching ends. The ends of 301 stitching shall be overstitched not less than 1 inch except where ends are turned under in a hem, held down by other stitching or where stitching is performed automatically. Where 301 stitching is performed automatically on stitch patterns such as box, box with cross-stitch, "W" stitching or straight line tacking, not less than three tying, overlapping , or back stitches shall be used to secure the ends of stitching.

3.3.1.3 Thread breaks. Thread breaks in stitching shall be overstitched not less than 1 inch at each break on stitch type 301, and not less than 1 ½ inches at each break on stitch type 401. Thread breaks noted during inspection shall be repaired by overstitching the existing stitching and the defective area for not less than one inch on type 301 stitching and 1 ½ inches on type 401 stitching on each side of the break. The ends of repair stitching are not required to be backstitched. Thread breaks in type 401 stitching may be overstitched with stitch type 301.

*3.3.1.4 Thread lubricant. The addition of any lubricant to the polyester sewing thread prior to or during the sewing operation is prohibited.

3.3.1.5 Skipped stitches. Two or more consecutively skipped stitches occurring in type 301 stitching shall be overstitched not less than 1 inch on each side of skips. Any skipped stitches in type 401 stitching shall be overstitched not less than 1-½ inches on each side of skip. Skipped stitches in 401 stitching may be overstitched with type 301 stitching.

3.3.1.6 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements of the stitch pattern, stitches per inch, size and type of thread are met and at least three or more tying, overlapping or backstitches are used to secure the ends of the stitching.

*3.4 Repairs. All necessary repairs shall be effected in accordance with Department of The Army Field Manual FM 10-16, except that hand-sewn patches, or cemented patches other than as specified herein will not be acceptable. Cemented patches on holes, cuts or tears not more than 1 inch in diameter or length, using cement conforming to Minnesota Mining and Manufacturing Co., part # 847 or equal (see 6.6) will be permitted. A maximum of two cemented patches per section will be allowable without prior contracting officer authorization. All other repairs shall require authorization by the contracting officer.

3.5 Location marks. Marks for showing location of component parts may be done with marking ink or by the use of drill holes. Ink marks or drill holes shall be located such that the marks or holes are completely and permanently covered by the component part.

3.6 Replacement of defective components. During the manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective or properly matched components.

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3.7 Marking. All labels and markings shall conform to DDD-L-20.

3.7.1 Identification marking. The letters "US" on the screen and cover and the nomenclature on the cover shall conform to type IV, class 9 of DDD-L-20, in the size characters and the locations shown on the drawings. Class 9 markings shall be clearly legible after exposure to accelerated weathering.

3.7.2 Identification labels. Identification labels shall conform to type VI, Class 6 of DDD-L-20 and shall be 3 ½ inches by 5 ½ inches minimum with a blank space of at least 1 ¼ inches provided on the bottom of the label for the Department of Defense Inspector's stamp. Identification labels shall be stitched to the screen and cover in the locations shown on the drawings.

3.7.3 Erection instruction labels. Labels of erection instructions shall conform to type VI, class 7 and shall be stitched to the screen and cover in the locations shown on the drawings. The label shall be a proportionate copy of Drawing 5-4-823 as reproduced from a glossy print of the drawing furnished by the contracting officer. The size label shall be, borderline to borderline, 15 inches ± ¼ inch in the long dimension, with a minimum of ½ inch margin outside of the borderline on all sides.

*3.8 Workmanship. The finished latrine screens shall conform to the quality of product established by this document.

4. VERIFICATION

4.1 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all dimensions specified have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2) it shall be examined for the defects listed in 4.4.3 and 4.4.4 and tested as specified in 4.4.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Conformance inspection. Sampling for inspection shall be as specified in the contract or purchase order.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 In-process examination. Inspection shall be made to determine that there is no lubrication of the polyester thread prior to or during the sewing operations as specified in 3.3.1.4.

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*4.4.3 End item visual examination. The end item shall be examined for the defects listed in Table I. The lot size shall be expressed in units of latrine screens or covers. The sample unit shall be one latrine screen or cover. The inspection level shall be I.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
Cloth	Cut, or tear longer than 1/8 inch; hole (including exposed drill hole) larger than 1/8 inch diameter:			
	- on screen		201	
	- on cover			301
	Exposed drill hole (1/8 inch or less) i.e., not Covered by attached parts:			
	- more than one but not more than four			302
	- more than four		202	
	Broken or missing yarn; multiple floats clearly visible at normal inspection distance (approximately 3 feet)		203	
Rope components	Any cut; any break			
	- on guy line		204	
	- on tie line			303
	Eye splice omitted		205	
	Eye splice not as specified			304
	Free ends not finished as specified			305
Hardware	Broken	101		
	Malformed:			
	- fails to perform intended function	102		
	- but will perform intended function		206	
	Corroded area			306
	Burr or sharp edge which may cause injury in handling or damage to fabric	103		
Poles & pins	Any pole or pin not as specified or omitted	104		
Grommets	Improperly set, two or more teeth exposed			307
	Clinched loosely, allowing grommet to rotate around hole			308
	Clinched excessively tight, cutting cloth	105		
	Insecurely clinched; i.e., grommets or washers become disengaged	106		
Seams & stitching	Open seam on stitch type 301:			
	- for more than ¼ inch but not more than 1inch			309
	- for more than 1 inch		207	

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TABLE I. End item visual defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
	Open seam on stitch type 401:			
	- one row of stitching open for 1 inch or less			310
	- one row of stitching open for more than 1 inch		208	
	- both rows of stitching open (not necessarily at the same location)	107		
	Note: A seam shall be classified as open when one or more stitches joining a seam are broken, or when one or more skipped or run-off stitches occur. On double stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.			
	Raw edges caught in stitching:			
	- more than 1 inch but not more than 3 inches			311
	- more than 3 inches		209	
	Thread breaks:			
	- overstitched less than 1 ½ inches on stitch type 401			312
	- overstitched less than 1 inch on stitch type 301		210	
	Note: Thread breaks not overstitched shall be classified as open seams.			
	Skipped stitches:			
	- overstitched less than 1 ½ inches on stitch type 401		211	
	- overstitched less than 1 inch on stitch type 301		212	
	Note: Skipped stitches not overstitched shall be classified as open seams.			
	Needle chews resulting in cut or tear larger than 1/8 inch in length, or hole larger than 1/8 inch in diameter:			
	- on screen		213	
	- on cover			313
	One or more required rows of stitching omitted	108		

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TABLE I. End item visual defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
Stitching ends	Overstitched less than 1 inch on stitch type 301 (except where ends are held down by other stitching, turned under in a hem, or where stitching is performed automatically)		214	
Seams	Pleated			
	- on screen		215	
	- on cover			314
Seam type	Wrong seam type	109		
Stitch type	Wrong stitch type	110		
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches		216	
	Tight, as evidenced by puckering on fabric, for more than 6 inches		217	
Stitches per inch	Less than the minimum specified:			
	- one stitch			315
	- two or more stitches		218	
	More than the maximum specified:			
	- one stitch			316
	- two or more stitches		219	
Note:	Variation in the number of stitches per inch caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or in turning corners shall be classified as follows:			
	a. With the minor "B" defect classification: - no defect			
	b. With the minor "A" defect classification: - minor "B" defect			
Stitching gage	Not as specified		220	
NOTE:	Applicable when double needle machine is used. When two rows of stitching are applied with a single needle machine, tolerances for stitching margin shall apply.			
Stitching margin	Larger than specified by more than 3/16 inch:			
	- for more than 2 inches but not more than 4 inches in length			317
	- for more than 4 inches in length		221	

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TABLE I. End item visual defects (continued)

Examine	Defect	Classification Major	Minor A	Minor B
Stitching Margin	Less than specified:			
	- for more than 2 inches but not more than 4 inches in length			318
	- for more than 4 inches in length		222	
Stitching indicated by "Z"on drawings	Chain portion of stitching shows on outside of screen or cover		223	
Mends, darns or patches	Repairs not in accordance with Department of the Army Field Manual FM 10-16 or not authorized by the contracting officer (see 3.4)	111		
Components & assembly	Any component or part omitted, not assembled or fabricated of applicable reference material	112		
Hems	Constructed with partial turnunder when double turnunder is required			319
	Twisted or pleated			320
Reinforce-ments	Improperly applied, causing excessive fullness on reinforcement or reinforced parts			321
Cementing, general	Puckers, loose edges, unadhered pocket or tunnels		224	
	Excess adhesion resulting in sags or lumps evident through outer layer of fabric		225	
	Adhesive exposed beyond the overcoated area on interior surface			322
General	Any other construction detail not performed as specified		226	
Marking & labels	Omitted, incorrect, illegible, or misplaced			323
Cleanness	Grease or oil stains			324
	Thread ends not trimmed to ½ inch or less			325

*4.4.4 End item dimensional examination. The end item shall be examined for conformance to the specified dimensions. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of shields. The sample unit shall be one shield. The inspection level shall be S-3.

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TABLE II. End item dimensional defects

Examine	Defect	Classification Major	Minor A	Minor B
Overall dimensions	Smaller than specified dimensions less applicable minus tolerance indicated on drawings:			
	- on screen	101		
	- on cover		201	
	Larger than specified dimensions and applicable plus tolerance indicated on drawings:			
	- on screen			301
	- on cover			302
Hems and seams	Hems or seams required to finish:			
	- ½ inch or more but less than 1-1/2 inches in width, less than specified by more than 1/8 inch		202	
	- 1 ½ inches or more but less than 2-3/4 inches in width, less than specified by more than 1/4 inch			303
	- 2 ¾ inches or more in width, more or less than specified by 1/2 inch			304
Rope components	Guy line less than specified length:			
	- by not more than 4 inches			305
	- by more than 4 inches but not more than 12 inches		203	
	- by more than 12 inches	102		
	Tie line less than specified finished length:			
	- by not more than 2 inches			306
	- by more than 2 inches but not more than 6 inches		204	
	- by more than 6 inches	103		
Grommets	Set beyond specified location by more than ½ inch (unless otherwise specified)			307

4.4.6 Packaging examination. The fully packaged end items shall be examined for marking, materials, workmanship and content defects.

5. PACKAGING

*5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the

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managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

*(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The latrine screens are intended for issue to units in the field for use at an outdoor latrine.

*6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is required (see 3.1, 4.2 and 6.4).
d. Levels of preservation and packing (see 5.1).

6.3 Sample. For access to standard samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Components. The following components, to be issued in the field, are required for the erection of the latrine screen.

Part	Quantity	Document
Pole, tent, upright, 7-foot	7	MIL-P-549
Pole, tent, ridge, 9-foot	2	MIL-P-549
Pole, tent, ridge, 7-foot	1	MIL-P-549
Pin, tent, wood, 16 inch	16	MIL-P-2383

*6.6 Equal item. Prior to the use of an “or equal” item, the supplier should submit the item, with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

*6.7 Subject term (key word) listing.
Cover
Mesh

*6.8 Changes from previous issue. The margins of this document have been marked with an asterisk(*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no

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liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Air Force – 82

Review activities:
Army – MD

Preparing activity:
DLA-CT

Project No: 8340-0600

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7 and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-43176D	**2. DOCUMENT DATE** ***(YYYYMMDD)*** 2000/SEP/15

3. DOCUMENT TITLE

SCREEN, LATRINE, FIRE, WATER WEATHER AND MILDEW RESISTANT TREATED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER PHILADELPHIA DSCP-CNRP	b. TELEPHONE *(Include Area Code)* (1) Commercial (215) 737-3290 (2) DSN 444-3290
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave. (Bldg 6, C&T) PHILADELPHIA, PA 19111-5092	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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